

BUCIN, RomaniaWMO: **151480**Lat: **46.6489N**Lon: **25.2964E**Elev: **4212**StdP: **12.59**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	5.9	-4.6	5.0	4.5	-0.7	6.2	9.7	16.6	16.2	14.0	19.6	4.5	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	75.1	61.5	72.2	59.9	69.7	58.7	62.8	71.8	61.4	69.6	60.1	67.7	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	90.5	64.5	58.5	85.6	64.1	57.0	80.9	63.6	30.6	72.0	29.5	69.7	28.5	68.1	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.1	9.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	21.8	25.5	29.5	39.5	49.7	56.2	57.8	59.2	50.7	42.2	35.6	25.1
	DBStd	15.23	9.04	8.34	8.22	9.44	6.29	5.84	5.62	6.05	8.87	8.44	8.56	8.39
	HDD50	4211	873	685	637	331	82	14	6	7	106	267	432	771
	HDD65	8735	1338	1105	1101	766	476	267	230	195	433	707	881	1236
	CDD50	979	0	0	0	15	71	201	249	291	126	25	1	0
	CDD65	28	0	0	0	0	0	4	7	13	4	0	0	0
	CDH74	135	0	0	0	0	0	18	28	71	18	0	0	0
	CDH80	4	0	0	0	0	0	1	0	3	0	0	0	0
Wind	WSAvg	3.7	3.6	3.9	4.2	3.9	3.6	3.6	3.6	3.5	3.6	3.5	3.1	3.8
Precipitation	PrecAvg	28.3	1.2	1.3	1.7	2.4	3.2	4.0	3.7	2.8	2.4	2.2	1.7	1.9
	PrecMax	37.8	2.2	3.0	4.1	4.3	5.6	7.0	6.4	4.7	5.8	4.4	3.8	6.2
	PrecMin	20.4	0.5	0.3	0.2	0.9	0.9	1.0	1.3	0.9	0.2	0.0	0.0	0.5
	PrecStd	4.3	0.6	0.7	0.9	1.0	1.2	1.4	1.5	1.0	1.4	1.2	0.8	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	46.0	54.8	65.7	70.1	75.9	77.2	79.4	77.2	67.3	57.5	46.1
		MCWB	36.1	39.1	43.3	50.6	54.2	61.7	62.2	62.3	62.4	52.9	45.6	37.3
	2%	DB	38.5	42.9	48.4	61.7	66.4	72.0	73.8	75.6	70.8	63.2	54.1	42.2
		MCWB	34.7	37.1	39.3	48.7	53.9	60.4	61.5	61.4	58.1	51.6	44.4	36.7
	5%	DB	35.9	39.6	44.3	57.6	63.2	69.5	70.8	72.3	66.9	58.7	50.8	38.9
		MCWB	33.4	35.0	37.1	47.3	53.1	59.3	59.9	60.0	56.1	49.7	43.3	35.1
	10%	DB	33.1	36.6	40.3	53.0	60.1	66.3	67.8	69.4	63.7	54.1	47.2	36.2
		MCWB	31.2	33.7	35.2	44.8	51.7	57.6	58.4	59.1	54.9	48.2	41.6	33.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	42.1	44.4	51.7	57.2	64.6	63.8	64.8	62.6	55.5	48.6	40.6
		MCDB	40.2	44.7	55.4	62.5	65.8	71.6	72.8	75.2	76.5	63.4	55.2	43.3
	2%	WB	36.0	37.6	40.6	49.9	55.2	61.8	62.3	62.8	59.2	52.9	46.4	38.3
		MCDB	38.0	40.4	46.8	59.4	62.7	69.1	71.0	71.8	68.4	59.9	51.3	40.6
	5%	WB	33.7	35.7	38.4	48.3	53.9	60.3	60.9	61.2	56.9	50.8	44.6	36.1
		MCDB	35.6	38.6	43.5	56.8	61.3	67.2	68.6	69.9	65.0	57.9	49.1	38.7
	10%	WB	31.3	33.9	36.0	45.7	52.4	58.8	59.5	60.0	55.2	48.4	42.6	33.6
		MCDB	33.0	36.2	39.5	52.3	59.2	65.6	66.3	68.1	62.5	53.5	46.4	35.6
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.3	12.5	14.4	14.2	14.9	14.4	15.2	13.0	13.3	10.7	10.0
		MCDBR	12.2	14.4	17.8	20.1	18.3	18.6	18.0	18.6	18.7	19.9	15.7	12.4
		MCWBR	9.0	10.0	11.1	10.9	10.0	9.6	9.7	9.3	9.4	12.1	10.1	8.3
	5% WB	MCDDBR	10.7	11.6	15.6	18.1	17.2	17.0	16.5	17.0	16.6	18.3	13.4	11.1
MCWBR		8.6	9.2	10.1	10.4	9.9	9.7	9.5	9.3	9.3	11.5	9.1	8.2	
Clear-Sky Solar Irradiance	taub	0.272	0.298	0.337	0.385	0.367	0.364	0.382	0.387	0.351	0.320	0.299	0.275	
	taud	2.372	2.330	2.311	2.219	2.329	2.359	2.307	2.291	2.372	2.458	2.464	2.418	
	Ebn at Noon	263	277	281	275	285	285	278	272	273	266	251	249	
	Edh at Noon	23	29	35	42	39	38	39	38	32	25	21	19	
All-Sky Solar Radiation	RadAvg	406	676	1027	1375	1699	1835	1818	1673	1191	814	458	308	
	RadStd	55	74	122	136	157	154	155	150	137	101	59	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (70 neighbors)	+1.26	N/A	N/A	+1.53	N/A	N/A	-275	-387	+157	+41

Nomenclature: See separate page